

Overfitting and Underfitting

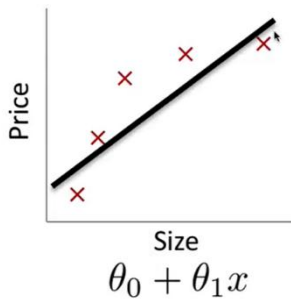
Model assessment: Estimating the prediction error of the model on **new** data.

Model selection: Choose the model with the best performance (i.e. the model with the smallest prediction error on new data).

Outlines:

1. Underfitting and overfitting
2. Training error vs. Test error
3. The bias-variance tradeoff
4. Estimating test error:
 - Validation and cross-validation
5. Model selection: Regularization

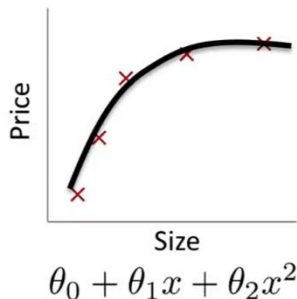
Example: Linear regression (housing prices)



Underfitting

High bias
Low variance

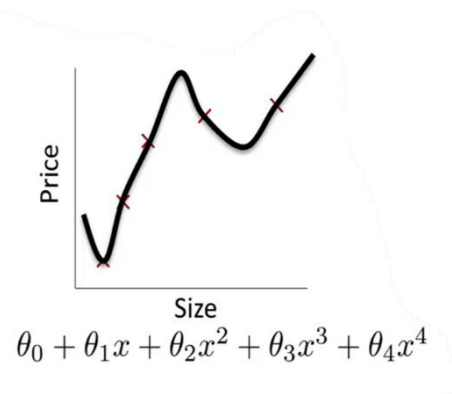
High interpretability
Low flexibility



"About right"

Bias
Variance

Interpretability
Flexibility



Overfitting

Low bias
High variance

Lower interpretability
Higher flexibility

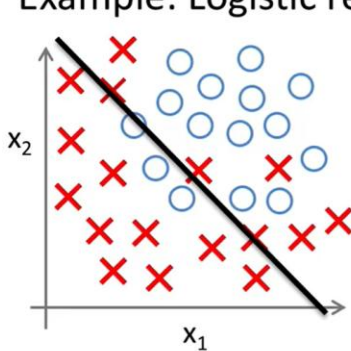
Underfitting

- Underfitting occurs when the model has too few free parameters, so it cannot fit the data adequately.
- The model will **NOT fit the training data well** and it will **FAIL to generalize** to new examples, since it is too simple for the problem.

Overfitting

- If we have too many features, the learned hypothesis may **fit the training set very well** (training error will be low), but **FAIL to generalize** to new examples (test error will be much higher).
- This occurs because the model is too flexible and adapts too closely to the noise in the particular training data used to train it.
- Training error does **NOT HAVE** to be zero, necessarily.

Example: Logistic regression



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

(g = sigmoid function)

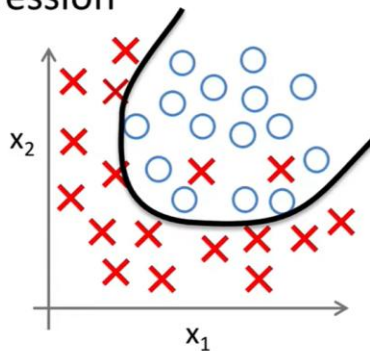
Underfit

High bias

Low variance

High interpretability

Low flexibility



$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$$

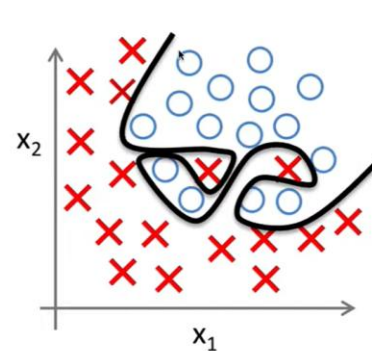
"About right"

Bias

Variance

Interpretability

Flexibility



$$g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \dots)$$

Overfit

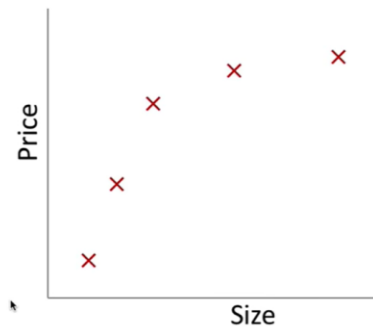
Low bias

High variance

Lower interpretability

Higher flexibility

x_1 = size of house
 x_2 = no. of bedrooms
 x_3 = no. of floors
 x_4 = age of house
 x_5 = average income in neighborhood
 x_6 = kitchen size
 $\vdots$
 x_{100}



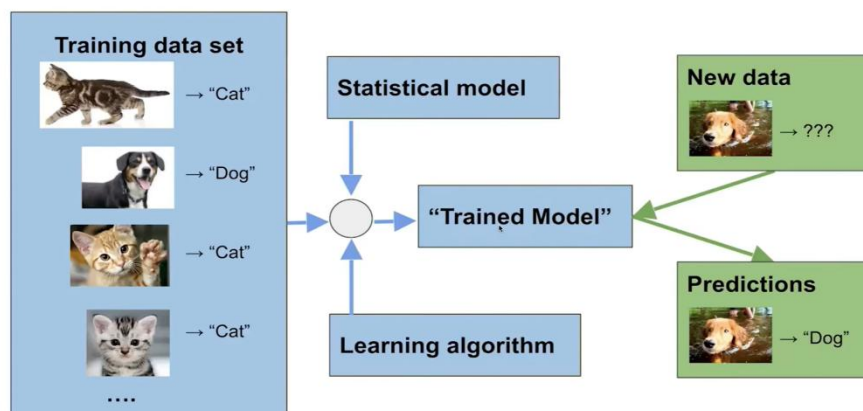
Training error vs Test error

The **training error** is the average error that results from using a machine-learning method to predict the response on **the training set**, i.e. the set of observations that was used to train the method. (Easy to calculate)

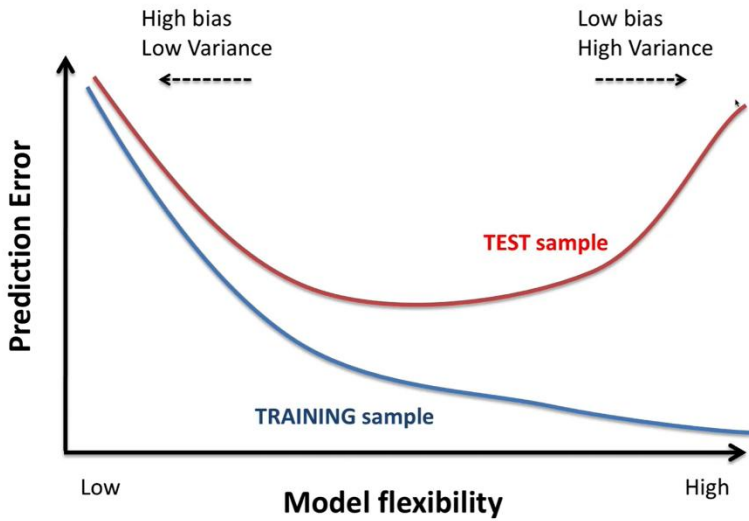
In contrast, the **test error** is the average error that results from using a machine-learning method to predict the response on a **new observation**, one that was not used in training the method. (Difficult to calculate, why?)

Training error is NOT a good estimate of the test error.

- Training error consistently decreases with model flexibility, typically dropping to zero if we increase the model flexibility enough.
- However, a model with zero training error is overfitting the training data and will typically generalize poorly.



TRAINING-SET PERFORMANCE vs. TEST-SET PERFORMANCE

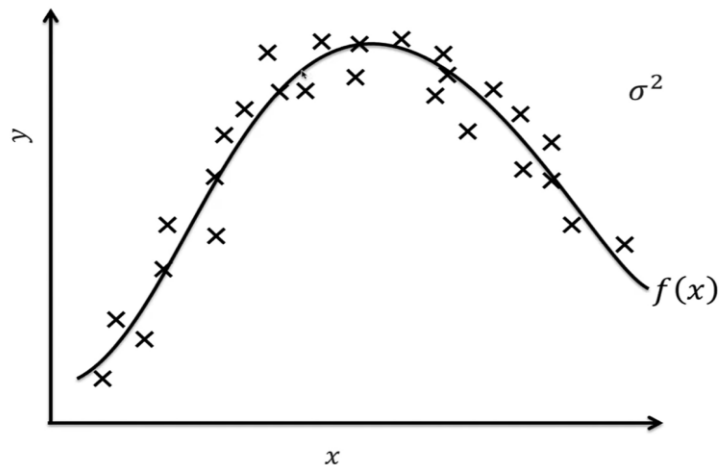


The bias-variance tradeoff

Bias-variance Decomposition:

We assume that $Y = f(X) + \varepsilon$, where $E[\varepsilon] = 0$ and $\text{Var}[\varepsilon] = \sigma^2$. We can derive an expression for the expected prediction error of a regression fit $\hat{f}(X)$ at an input point $X = x_0$, using squared-error loss:

$$\text{Err}(x_0) = E \left[\left(Y - \hat{f}(x_0) \right)^2 \mid X = x_0 \right]$$



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We can derive an expression for the expected prediction error of a regression fit $\hat{f}(X)$ at an input point $X = x_0$, using squared-error loss:

$$\begin{aligned} \text{Err}(x_0) &= E \left[\left(Y - \hat{f}(x_0) \right)^2 \mid X = x_0 \right] = \\ &\sigma^2 + (E[\hat{f}(x_0)] - f(x_0))^2 + E \left[\hat{f}(x_0) - E[\hat{f}(x_0)] \right]^2 = \\ &\sigma^2 + \text{Bias}^2(\hat{f}(x_0)) + \text{Variance}(\hat{f}(x_0)) \end{aligned}$$

Irreducible Error: variance of the target (Y) around its true mean $f(x_0)$. This cannot be avoided no matter how well we estimate $f(x_0)$. (Unless $\sigma^2 = 0$, of course.)

Due to the noise ε .

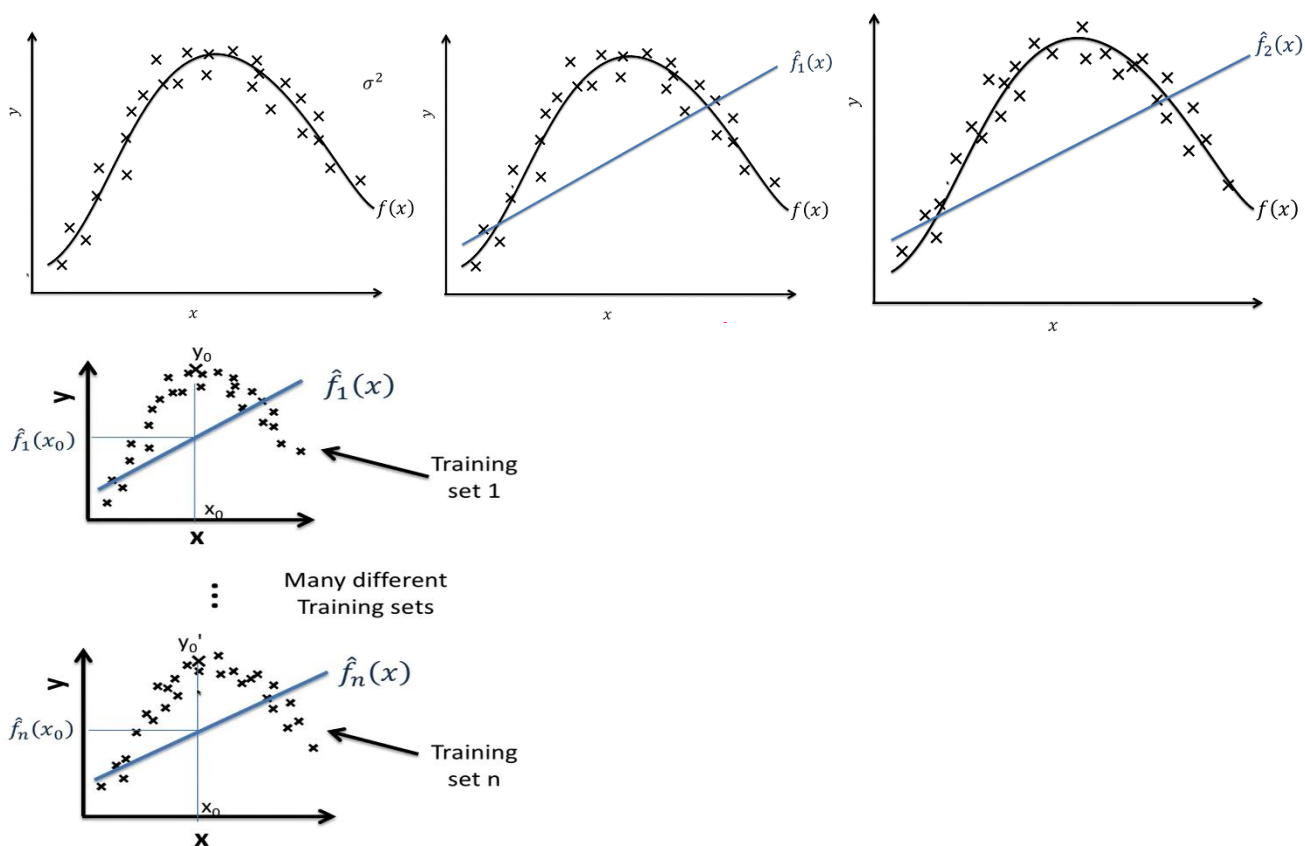
Bias: the amount by which the average of our estimates differs from the true mean.

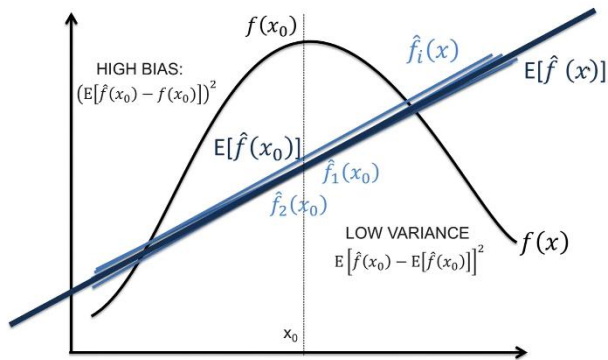
Mainly due to having a structure too rigid (model too simple), so our model $\hat{f}$ systematically deviates from the true function f .

Variance: the expected squared deviation of $\hat{f}(x_0)$ around its mean.

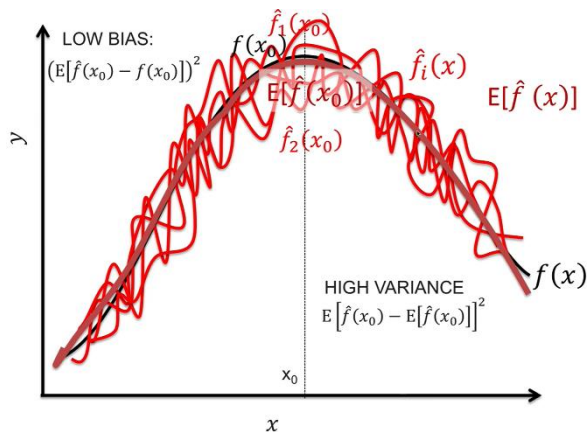
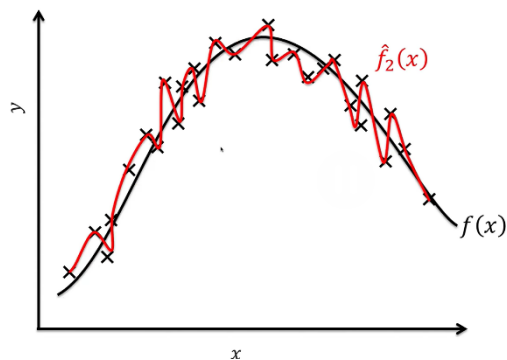
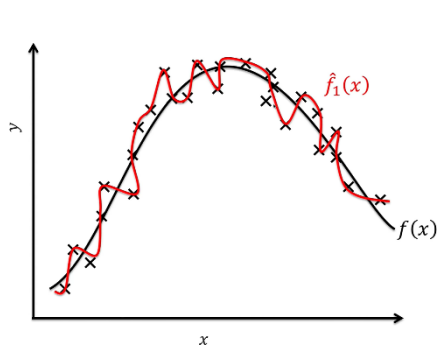
Mainly due to sampling and to having a structure too flexible, so the estimated model $\hat{f}$ changes a lot with different training samples.

Typically the more complex we make the model $\hat{f}$, the lower the (squared) bias but the higher the variance.





For Another higher degree model:



The Bias-Variance Tradeoff

In machine learning, the **bias-variance dilemma** or **bias-variance tradeoff** is the problem of simultaneously minimizing:

- **The bias:** how far off on average—ACROSS TRAINING SETS—the estimate is from the truth.
- **The variance:** how much the estimate varies around its average—ACROSS TRAINING SETS.

This tradeoff applies to all forms of supervised learning: regression, classification, function fitting, decision trees, neural networks, etc.

The bias-variance tradeoff is a central problem in supervised learning.

- **Models with high bias:** These are intuitively simple models that impose restrictions on the kind of regularities that can be learned (examples include linear classifiers). The problem with these models is that they **underfit**, i.e. not learn the relationship between predicted (target) variables and features.
- **Models with high variance:** These are those that can learn many kinds of complex regularities—but, because of this, they can easily learn noise in the training data, i.e. **overfitting**.

To achieve good performance on data outside the training set, a tradeoff must be made.